Citi Bike: Demand, Docking Costs & Transportation Access

Revised machine-learning project and Databricks implementation plan
CUNY 622 Final Project | October 1, 2026

Revised research question

How can Citi Bike allocate docking-station investment to improve utilization and transportation access within a limited budget?

The proposed project combines station-level activity forecasting with dock-capacity analysis, estimated financial costs, and neighborhood transportation needs. Forecasts support cost scenarios; access and reliability measures help interpret whether a station needs more capacity, better operations, or further investigation.

Recommended scope

Start with one complete recent 12-month period. Predict hourly departures and arrivals, summarize station-month use relative to capacity, estimate annual station costs under low/base/high assumptions, and map access gaps. Add empty/full-station analysis only where timestamped availability data exist.

Component	Revised purpose
Machine learning	Predict station departures and arrivals using history, calendar, weather, and location.
Docking-station use	Compare throughput per dock and identify temporal imbalances.
Financial assessment	Estimate cost per departure and test capital and operating-cost assumptions.
Social impact	Evaluate access, reliability, and model errors across neighborhoods.
Decision support	Compare investment scenarios under a budget with explicit equity criteria.

What changes from the original plan

The Bronze/Silver/Gold trip pipeline remains. Add effective-dated station capacity, availability snapshots, annual program financials, documented cost assumptions, and station-month performance tables. Financial calculations are a scenario-analysis layer, not a separate requirement to predict actual station profits.

Evidence note: Sample statistics and source benchmarks below are carried forward from the preceding analysis. The full Mac archive has not been inventoried in this session; all study months and cost inputs must be validated before modeling.

1. Data foundation and additional sources

Sample finding	December 2013	August 2026, part 1
Trip rows	443,966	1,000,000
Actual start-date coverage	December 1-31	July 31-August 14
Starting stations	329	2,308
Median duration	8.9 minutes	9.6 minutes
Rider/bike fields	Birth year, gender, bike ID	Ride ID; electric/classic type
Membership	Subscriber / Customer	Member / Casual

The modern attachment is a partial release, not a complete month. Include every part before computing monthly totals. System expansion, seasonality, electric-bike adoption, and schema changes make a direct 2013-versus-2026 comparison unsuitable as the main model design.

Prioritized external data

Dataset	Join and purpose
Citi Bike station information / GBFS	Station ID and snapshot time: capacity, availability, disabled equipment, operating status. Archive prospectively; a current feed does not reconstruct historical status.
NOAA hourly weather	Time and nearby weather station: temperature, precipitation, wind. Operational forecasts need weather available at prediction time.
ACS five-year estimates + tract boundaries	Spatially assign station locations to tracts. Use vehicle availability, poverty, population, and commuting characteristics; retain margins of error and geography vintage.
MTA station locations / hourly ridership	Transit proximity; lagged ridership where available. Do not use unpublished future counts.
NYCHA locations	Assess nearby public-housing access and walking distances.
NYC DOT / IBO financial reports	Annual program costs and scenario installation benchmarks; preserve reporting scope and dollar year.

Optional extensions include NYPD cyclist-injury collisions and NYC bicycle routes for safety analysis. Trip endpoints do not establish actual cycling routes or total bicycle exposure.

Important limits

Completed trips measure observed use, not all unmet demand. Neighborhood demographics describe the station surroundings, not rider residence or identity. Missing station IDs, implausible durations, and legacy values such as \N require explicit quality flags.

2. Dock-capacity and service-use measures

Create station-hour activity for forecasting and station-month summaries for capacity and cost assessment. Keep departures and arrivals separate; their sum measures station transactions and counts each trip twice across the network.

Measure	Definition / interpretation
Departures per dock per day	Period departures / sum of daily dock capacity. Use fractional dock-days if capacity changes within a day; restrict to the station operating period.
Arrivals per dock per day	Period arrivals / dock-days. Describes return throughput separately from departures.
Net flow	Departures minus arrivals by hour. Indicates imbalance pressure; it does not measure actual rebalancing.
Empty-station share	Observed rental-enabled time with no rentable bikes / observed rental-enabled time. Report disabled-service time separately.
Full-station share	Observed return-enabled time with no usable return docks / observed return-enabled time.
Availability coverage	Observed valid snapshot minutes / expected monitoring minutes. Report gaps and avoid interpolating across long missing intervals.

Historical station capacity

Build a station-location and capacity history with station ID, coordinates, effective start/end times, capacity, operating status, source, and confidence. Do not apply today's dock count to historical trips. Where historical capacity is unavailable, restrict the capacity analysis to supported dates or clearly label a constant-capacity sensitivity scenario.

Interpreting utilization

High departures per dock can reflect strong demand or an undersized station. Low departures can reflect low demand, lack of bikes, poor connectivity, disabled service, or station deployment bias. Throughput is not dock occupancy; occupancy and service-failure measures require status snapshots.

Potential interventions

Option	Evidence needed
Add docks	Capacity constraints and return failures, credible marginal cost, and a stated additional-use assumption.
Improve rebalancing / maintenance	Repeated empty/full periods or disabled equipment; operating cost estimates.
Add a station / strengthen connections	Access gaps, neighborhood need, network links, candidate-site feasibility.
Investigate a low-use station	Availability, active dates, nearby alternatives, and barriers to use before drawing removal conclusions.

3. Financial assumptions and investment scenarios

Treat financial results as estimated cost efficiency. Trip files cannot reconstruct exact fares, membership revenue, or sponsorship allocation, and public program totals do not supply station-level accounts.

Public benchmarks carried forward

The preceding analysis identified NYC DOT's 2025 financial report as reporting approximately $228.1 million in revenue, $203.4 million in operating expenses, and $24.6 million in operating income. Its reporting scope includes Jersey City/Hoboken; do not allocate those full totals to an NYC-only station sample.

The NYC Independent Budget Office's 2025 analysis was identified as estimating roughly $85,000 for installation of a 27-dock station in 2024 dollars, based on an inflation-adjusted San Francisco comparison. Use this as a sourced scenario benchmark, not an actual NYC station invoice or a linear price for every additional dock. Links appear in the source register.

Core cost calculation

```
Estimated annual cost per departure =
  (annualized station capital cost
   + direct station operating cost
   + allocated shared operating cost)
  / annual station departures
```

For a simple undiscounted scenario, annualized capital cost equals (installed cost minus residual value) divided by useful life. If a discount rate is used, apply a consistent equivalent-annual-cost method. Do not add depreciation for the same assets on top of annualized capital costs.

Parameter	Required treatment
Capital	Installation, replacement, electrification, site work, useful life, residual value, dollar year.
Direct operations	Maintenance, station-specific field work and rebalancing, where observed; otherwise explicit estimates.
Shared costs	Allocate only the relevant expense pool. Compare dock-count, activity, and mixed allocation rules.
Scenario uncertainty	Low/base/high costs, useful life, allocation method, and forecast activity; preserve source and version.
Revenue (optional)	Document fare and shared-revenue allocation assumptions. Label estimated revenue separately from measured activity.

Budget comparison

For each proposed intervention, show incremental annualized cost, assumed additional departures, access improvements, and reliability benefit. Cost per additional departure is meaningful only when incremental use is positive and defensibly estimated. Baseline forecasts alone cannot establish the causal effect of new docks.

Present a Pareto comparison of efficiency and access, or use an explicit minimum-access requirement. Any weighted score should disclose its weights and show how rankings change under alternative choices.

4. Machine-learning design and evaluation

Target and feature table

Use one row per operating station per hour. Model departures and arrivals separately. Include zero-trip hours for stations known to be active; when status is unavailable, document inferred active periods and test sensitivity.

Features: hour, weekday, season, holidays, prior-hour and prior-week activity, lagged rolling averages, weather, station location, and nearby transit. Add historically valid capacity where supported. Use neighborhood characteristics for equity analysis and optionally compare a separate model that includes them.

Model	Role
Same station/hour, prior week	Operational baseline; specify a fallback for new stations and missing history.
Poisson regression	Interpretable count baseline; consider negative binomial if overdispersion warrants it.
Gradient-boosted trees	Capture nonlinear demand patterns and interactions; use an appropriate count objective or nonnegative prediction treatment.

Validation

Split chronologically into training, validation, and final test periods; use rolling-origin validation where feasible. Build lagged features without future information. Evaluate MAE and count deviance and report results by busy/quiet stations, neighborhood groups, and capacity categories. A geographic holdout separately tests transfer to unfamiliar stations.

For an operational forecast, weather and transit features must be available at forecast time. A model using future observed weather is a retrospective experiment and must be labeled accordingly. Future multi-step forecasts cannot use realized intervening trip counts.

Financial and social assessment

Aggregate predictions to station-month and annual scenarios. Report activity uncertainty and sensitivity to cost assumptions, rather than presenting a single precise financial ranking. Keep projected cost efficiency separate from causal estimates of investment benefits.

Dimension	Final output
Predictive performance	Baseline/model comparison and error breakdowns.
Efficiency	Capacity-adjusted throughput and estimated cost per departure.
Reliability	Empty/full periods and snapshot coverage, if available.
Access	Station proximity for households without cars, public housing, and transit connections.
Investment	Scenario comparison under budget and equity requirements.

Low observed activity should not automatically reduce a neighborhood's investment priority. Inadequate availability or network coverage can suppress recorded use. Expansion modeling also inherits the bias of existing station placement.

5. Databricks import and processing plan

Local Python/Spark is suitable for inventory and prototyping. Databricks Connect supports local development while Spark runs remotely; the cloud cluster cannot read the Mac path directly. Pilot a few complete months in a Unity Catalog volume, then use private S3 in the workspace region for the full archive if needed.

1. Inventory and choose complete periods

On the Mac, inventory /Volumes/BigRaid/RandyFiles/Cuny/622/FinalProject. Record relative path, byte size, checksum, actual header/schema family, row count, nominal month/part, and timestamp range. Exclude duplicate ZIP/extracted copies; verify every selected month is complete.

2. Land immutable raw files

```
CREATE SCHEMA IF NOT EXISTS main.cuny622;
CREATE VOLUME IF NOT EXISTS main.cuny622.citibike_raw;
```

Run the following on the Mac after configuring the Databricks CLI. Adjust paths and catalog to the workspace and privileges.

```
databricks fs cp \
  "/Volumes/BigRaid/RandyFiles/Cuny/622/FinalProject/201312-citibike-tripdata.csv" \
  "dbfs:/Volumes/main/cuny622/citibike_raw/201312-citibike-tripdata.csv"
```

For S3, preview CSV-only synchronization, review the result, then remove --dryrun. Connect through a Unity Catalog IAM-backed storage credential and external location; keep credentials out of notebooks.

```
aws s3 sync \
  "/Volumes/BigRaid/RandyFiles/Cuny/622/FinalProject/" \
  "s3://YOUR-BUCKET/citibike/raw/incoming/" \
  --exclude "*" --include "*.csv" \
  --no-follow-symlinks --dryrun
```

3. Bronze: preserve raw evidence

Separate incompatible historical and modern schemas by actual headers. Read raw fields as strings; retain source filename, checksum, ingestion timestamp, schema version, and malformed records. Normalize column names for Delta. Make batch reruns idempotent; later Auto Loader ingestion needs separate checkpoints and schema locations for incompatible families.

4. Silver: validated, effective-dated records

Unify start/end timestamps and station IDs; keep IDs as strings, including values such as 6322.01. Map Subscriber/Customer to Member/Casual while preserving original labels. Retain unavailable bike/demographic fields as null. Parse fractional seconds, preserve local wall time, validate timezone conventions, and document daylight-saving ambiguity.

Flag missing endpoints, invalid coordinates, implausible durations, duplicate ride IDs, and suspect demographics. Use a separate historical duplicate policy. Normalize station capacity/status histories and preserve financial scope, dollar year, source, and assumption versions.

6. Gold tables, work sequence, and deliverables

Gold / supporting table	Contents
station_hour_activity / features	Departures, arrivals, zero hours, lagged activity, calendar, weather.
station_location_history / capacity_history	Effective-dated coordinates, station identity, docks, active status.
station_status_snapshots	Bikes/docks available, disabled equipment, operating flags, timestamp.
station_tract_crosswalk / access_indicators	Station-to-tract mapping, geography vintage, ACS and transit access.
program_financials / cost_assumptions	Annual expense categories and scope; sourced low/base/high inputs.
station_month_performance	Observed/forecast activity, dock-days, cost scenarios, access, reliability.
investment_scenarios	Intervention, incremental cost, assumed benefit, uncertainty, budget outcome.

Spatially join the small station-location history to tract polygons, then attach the crosswalk to trips. Aggregate trips before joining annual financial inputs to avoid duplicating costs across trip rows. Keep feature generation distributed and log data versions, assumptions, model parameters, and metrics with MLflow.

Stage	Concrete completion criterion
1. Inventory / pilot	Complete-month manifest, reconciled Bronze rows, validated Silver samples.
2. Assemble modeling data	Recent complete period; station-hour zeros/lags, weather and tract crosswalk.
3. Baselines and ML	Chronological evaluation, model comparison, neighborhood error analysis.
4. Capacity / costs	Supported historical capacity and versioned cost scenarios; sensitivity results.
5. Access / decisions	Access map, station comparisons, and documented investment scenarios.
6. Final submission	Reproducible notebooks, data dictionary, methods report, figures, limitations.

Quality and compute checks

Reconcile raw/imported counts and month coverage, check overlapping downloads, inspect station-hour totals and capacity dates, and report availability coverage. Confirm cost allocation reconciles to the relevant financial pool and avoids duplicate capital treatment. Start with modest compute and automatic termination where available.

Minimum viable final project

Deliver an hourly demand model, station-month capacity-adjusted use, estimated cost-per-departure scenarios, and an access-gap map. Availability monitoring, causal dock-expansion analysis, and rebalancing optimization remain optional extensions. This retains a measurable ML objective while adding a financial decision layer and a social-betterment purpose.

Source register

Links carried forward from the preceding analysis. Validate dataset versions, historical coverage, reporting scope, and cost benchmarks when building the final research dataset.

Citi Bike system data and GBFS

NYC DOT: 2025 bike-share financial data

NYC IBO: Citi Bike lessons for the future (2025)

Census ACS API

TIGERweb tract boundaries

NOAA Integrated Surface Database

MTA hourly subway ridership, beginning 2025

NYCHA Development Data Book

NYPD Motor Vehicle Collisions - Crashes

NYC Bicycle Routes

Databricks Connect

Unity Catalog volume files

Databricks CLI fs commands

S3 storage with Unity Catalog

Databricks Auto Loader

AWS CLI s3 sync

Prepared as an integrated revision of the original project plan, incorporating docking-station financial cost and use. No new empirical modeling or full-archive inspection was performed for this document.